RT Interact Phase 1

Expert Clinical Validation

Reviewer Pack — Phase 1

June 2026 · 42 Synthetic Validation Scenarios · CONFIDENTIAL — Not for external distribution

1. Purpose of This Review

RT Interact is a decision-support prototype that generates alerts for potential interactions between radiotherapy (RT) and systemic anticancer agents. This expert review is the first formal content-validity assessment of the prototype.

You are asked to assess, for each of 42 synthetic clinical scenarios, whether the app's current output — including alert level, toxicity domains, risk driver, rationale, and uncertainty signalling — is clinically appropriate. Your responses will be used to identify rule gaps, calibration errors, and false-reassurance risks before any further development.

Reviews are independent and blinded. Please complete your assessment without reference to other reviewers' opinions. A minimum of three Radiation Oncology specialist reviewers is required per scenario before adjudication.

2. Prototype Status and Disclaimer

IMPORTANT: RT Interact is a research and quality-improvement prototype. It is not approved or cleared for clinical reliance in any jurisdiction. It must not be used to guide patient care. All scenarios in this pack are entirely synthetic — no patient data is included.

Specific limitations relevant to this review include:

- Content has not been verified against eviQ cancer treatment protocols or the ARTG/PBS.
- Alert levels, toxicity domains, and rationale text have not undergone prior expert consensus validation.
- The tool does not account for patient-specific factors (renal/hepatic function, performance status, comorbidities).
- No management recommendations are provided.
- Combination-specific synergistic toxicity from multiple simultaneous agents is not modelled in Phase 1.

- Rules are based on categorical inputs only — continuous dose variables (Gy) are not modelled.

A full list of known limitations is provided in Section 7 of this document.

3. App Access Instructions

The RT Interact prototype is accessible via a private web link provided with this pack. No installation or login is required — the app runs entirely in your web browser.

Step 1: Open the private app link provided in your invitation.

Step 2: In the app, select the systemic agent from the dropdown list.

Step 3: Select the RT site and subsite.

Step 4: Select the timing of the systemic agent relative to RT.

Step 5: Select the fractionation category if applicable (some scenarios require this field).

Step 6: Review the app output: alert level, toxicity domains, risk driver, rationale text, and evidence references.

Step 7: Complete the online review form for this scenario (link provided in your invitation).

The app runs in-browser only. No patient information should be entered at any time.

4. Review Workflow

For each scenario, you will:

1. Read the scenario summary in this document (Section 6).
2. Enter the listed inputs into the RT Interact app.
3. Review all components of the app output.
4. Complete the online review form for that scenario.
5. Repeat for all assigned scenarios.

What you are assessing

For each scenario, the online form asks you to rate:

Alert appropriate: Is the app alert level clinically appropriate?

Preferred alert level: If not, what level is more appropriate?

Toxicity domains appropriate: Are the listed toxicity domains correct and complete?

Risk driver appropriate: Is the stated primary risk driver appropriate?

Rationale appropriate: Is the rationale narrative accurate and appropriately qualified?

Evidence appropriate: Is the evidence basis adequate for the alert level?

Uncertainty signalling appropriate: Is the degree of uncertainty appropriately communicated?

Potential false reassurance: Does the output risk creating false reassurance — i.e. under-alerting a real risk?

Reviewer confidence: How confident are you in your assessment of this scenario?

Suggested correction: Optional free-text correction or comment.

You are assessing content validity only — not interface design, technical implementation, or Phase 2 features (dose constraints, management recommendations).

Guidance on specific scenario types

Missing-input scenarios (S032–S034)

The app should return an incomplete or prompt state — not a clinical alert. Assess whether this handling is clearly communicated.

Fallback scenarios

These return 'Uncertain / evidence limited' because no matching rule was found. Assess: (a) Is Uncertain appropriate given the available evidence? (b) Are the listed toxicity domains clinically relevant? (c) Does Uncertain risk implying the combination is safer than it actually is?

Multi-agent scenarios (S035–S036)

Each agent is evaluated individually. Combination-specific synergistic effects are not modelled. Assess whether the individual-agent output is adequate or whether multi-agent interactions represent a clinically significant gap.

Critical false-reassurance scenarios

S024 (Gemcitabine + lung RT), S025 (Temozolomide + brain RT), and S036 (Carboplatin/paclitaxel + lung RT) are known or standard combinations that may currently return Uncertain due to rule gaps. Please assess these with particular care.

5. Alert Level Reference

The app uses five alert levels. Use the table below when completing the review form.

Alert Level	Clinical Meaning
No specific alert	No interaction signal for this combination; standard clinical checks apply.
Caution	Recognised combination with expected managed toxicity (e.g. protocol chemoradiation); awareness required.
Moderate toxicity alert	Meaningful evidence of enhanced toxicity requiring active monitoring.
High toxicity alert	Well-documented severe or life-threatening enhanced toxicity; specialist coordination required.
Uncertain / evidence limited	Insufficient evidence to assign an alert level; do not interpret as low risk; specialist review recommended.
Incomplete (S032–S034)	App returns a prompt or incomplete state when required inputs are missing.

6. Validation Scenarios

This section lists all 42 synthetic validation scenarios. Each scenario shows the inputs to enter into RT Interact and a brief task description. Do not interpret the absence of an expected answer as guidance — assess the app output independently based on your clinical judgement.

Scenario overview

Category	Scenarios	Focus
Low concern / No specific alert	S001–S005	Confirm no false-high alerts for standard low-risk sequential combinations.
Caution	S006–S008	Protocol chemoradiation and ICI + brain SRS; confirm appropriate calibration.
Moderate toxicity alert	S009–S017	Key combination alerts; assess toxicity domain and rationale adequacy.
High toxicity alert	S018–S023	Anti-VEGF + SBRT, CAR-T, bispecifics; confirm High alert is clinically warranted.
Uncertain / evidence limited	S024–S029	Known and novel agents with limited rules; assess false-reassurance risk.
Unmatched / evolving evidence	S030–S031	Novel agents returned via fallback; assess whether Uncertain is appropriate.
Missing input states	S032–S034	Incomplete inputs; confirm safe handling with no premature alert.
Multi-agent	S035–S036	Dual ICI and platinum+taxane doublet; assess combination gap adequacy.
Timing-sensitive	S037–S039	Anti-VEGF timing, recent capecitabine, anthracycline recall; assess timing logic.
Fractionation-sensitive	S040–S041	ICI + SBRT vs conventional; CDK4/6 domain change with fractionation.
Site-specific interaction	S042	Trastuzumab + breast RT; classification and alert assessment.

Low concern — No specific alert (S001–S005)

Scenario S001

Agent: Enzalutamide

Agent class: Androgen receptor antagonist / ADT

RT site: Prostate — Prostate only

Timing: Sequential

Fractionation: Palliative single fraction

Task: *Enter these inputs into RT Interact and assess whether the alert level is appropriate for this standard sequential low-risk combination.*

Scenario S002

Agent: Letrozole

Agent class: Aromatase inhibitor / endocrine therapy

RT site: Breast — Whole breast

Timing: Planned after RT

Fractionation: Palliative multifraction

Task: *Enter these inputs into RT Interact and assess whether the alert level is appropriate for this standard sequential low-risk combination.*

Scenario S003

Agent: Abiraterone

Agent class: Androgen synthesis inhibitor / ADT

RT site: Prostate — Prostate + pelvic nodes

Timing: Sequential

Fractionation: Palliative multifraction

Task: *Enter these inputs into RT Interact and assess whether the alert level is appropriate for this standard sequential low-risk combination.*

Scenario S004

Agent: Bicalutamide

Agent class: Androgen receptor antagonist / ADT

RT site: Spine — Lumbar/sacral

Timing: Sequential

Fractionation: Palliative multifraction

Task: *Enter these inputs into RT Interact and assess whether the alert level is appropriate for this standard sequential low-risk combination.*

Scenario S005

Agent: Anastrozole

Agent class: Aromatase inhibitor / endocrine therapy

RT site: Breast — Chest wall

Timing: Planned after RT

Fractionation: Palliative single fraction

Task: *Enter these inputs into RT Interact and assess whether the alert level is appropriate for this standard sequential low-risk combination.*

Caution (S006–S008)

Scenario S006

Agent: Paclitaxel

Agent class: Cytotoxic chemotherapy — taxane

RT site: Head and neck — Larynx/hypopharynx

Timing: Concurrent

Fractionation: Conventional fractionation

Treatment intent: Curative

Task: *Enter these inputs into RT Interact and assess whether the alert level and toxicity domains are clinically appropriate for this recognised protocol combination.*

Scenario S007

Agent: Ipilimumab

Agent class: Immune checkpoint inhibitor — anti-CTLA-4

RT site: Brain — SRS/focal

Timing: Concurrent

Fractionation: SRS

Task: *Enter these inputs into RT Interact and assess the alert level, toxicity domains, rationale, and evidence for clinical appropriateness.*

Scenario S008

Agent: Docetaxel

Agent class: Cytotoxic chemotherapy — taxane

RT site: Head and neck — Oral cavity/oropharynx

Timing: Concurrent

Fractionation: Conventional fractionation

Treatment intent: Curative

Task: *Enter these inputs into RT Interact and assess whether the alert level and toxicity domains are clinically appropriate for this recognised protocol combination.*

Moderate toxicity alert (S009–S017)

Scenario S009

Agent: Cisplatin

Agent class: Cytotoxic chemotherapy — platinum / strong mucosal radiosensitiser

RT site: Head and neck — Oral cavity/oropharynx

Timing: Concurrent

Fractionation: Conventional fractionation

Treatment intent: Curative

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S010

Agent: Palbociclib

Agent class: CDK4/6 inhibitor

RT site: Pelvis — Rectal

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S011

Agent: Osimertinib

Agent class: EGFR TKI (3rd generation)

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S012

Agent: Trastuzumab deruxtecan (T-DXd)

Agent class: HER2-directed antibody-drug conjugate

RT site: Lung — Central

Timing: Concurrent

Fractionation: Moderate hypofractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S013

Agent: Nivolumab

Agent class: Immune checkpoint inhibitor — anti-PD-1

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S014

Agent: Dabrafenib

Agent class: BRAF inhibitor

RT site: Skin/superficial — Skin lesion

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S015

Agent: Cetuximab

Agent class: EGFR-targeted monoclonal antibody

RT site: Oesophagus — Mid-thoracic

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S016

Agent: Ribociclib

Agent class: CDK4/6 inhibitor

RT site: Pelvis — Gynaecological

Timing: Concurrent

Fractionation: Moderate hypofractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

Scenario S017

Agent: Gemcitabine

Agent class: Cytotoxic chemotherapy — antimetabolite / gemcitabine

RT site: Spine — Thoracic

Timing: Concurrent

Fractionation: Large-field marrow-exposing RT

Task: *Enter these inputs into RT Interact and assess whether the alert level, toxicity domains, and rationale are clinically appropriate.*

High toxicity alert (S018–S023)

Scenario S018

Agent: Bevacizumab

Agent class: Anti-VEGF monoclonal antibody

RT site: Upper abdomen — Pancreas

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the alert level is clinically warranted and whether all relevant toxicity domains are captured.*

Scenario S019

Agent: Axitinib

Agent class: Anti-VEGFR TKI

RT site: Upper abdomen — Liver

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the alert level is clinically warranted and whether all relevant toxicity domains are captured.*

Scenario S020

Agent: Sorafenib

Agent class: Multi-kinase / anti-VEGFR inhibitor

RT site: Upper abdomen — Adrenal

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the alert level is clinically warranted and whether all relevant toxicity domains are captured.*

Scenario S021

Agent: Tisagenlecleucel

Agent class: CAR-T cell therapy

RT site: Brain — Whole brain

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level is clinically warranted and whether all relevant toxicity domains are captured.*

Scenario S022

Agent: Tarlatamab

Agent class: Bispecific T-cell engager (DLL3xCD3)

RT site: Head and neck — Oral cavity/oropharynx

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the alert level is clinically warranted and whether all relevant toxicity domains are captured.*

Scenario S023

Agent: Ramucirumab

Agent class: Anti-VEGFR2 monoclonal antibody

RT site: Oesophagus — Lower/GOJ

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the alert level is clinically warranted and whether all relevant toxicity domains are captured.*

Uncertain / evidence limited (S024–S029)

Scenario S024

Agent: Gemcitabine

Agent class: Cytotoxic chemotherapy — antimetabolite / gemcitabine

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is clinically appropriate, including whether there is any risk of false reassurance.*

Scenario S025

Agent: Temozolomide

Agent class: Alkylating-like agent — TMZ

RT site: Brain — Whole brain

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is clinically appropriate, including whether there is any risk of false reassurance.*

Scenario S026

Agent: Olaparib

Agent class: PARP inhibitor

RT site: Pelvis — Gynaecological

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is clinically appropriate, including whether there is any risk of false reassurance.*

Scenario S027

Agent: Venetoclax

Agent class: BCL-2 inhibitor

RT site: Pelvis — Rectal

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is clinically appropriate, including whether there is any risk of false reassurance.*

Scenario S028

Agent: Datopotamab deruxtecan

Agent class: TROP2-directed antibody-drug conjugate (novel)

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is clinically appropriate, including whether there is any risk of false reassurance.*

Scenario S029

Agent: Zolbetuximab

Agent class: Claudin 18.2 monoclonal antibody (novel)

RT site: Oesophagus — Lower/GOJ

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is clinically appropriate, including whether there is any risk of false reassurance.*

Unmatched / evolving evidence (S030–S031)

Scenario S030

Agent: Sotorasib

Agent class: KRAS G12C inhibitor (evolving evidence)

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the output is appropriate for this novel or evolving-evidence agent.*

Scenario S031

Agent: Clodronate

Agent class: Bone-modifying agent — first-generation bisphosphonate

RT site: Spine — Lumbar/sacral

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the output is appropriate for this novel or evolving-evidence agent.*

Missing input states (S032–S034)

Scenario S032

Agent: (No agent selected)

Agent class: —

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *In the RT Interact app, select the RT site and timing shown but leave the agent field empty. Assess how the app handles this incomplete input.*

Scenario S033

Agent: Gemcitabine

Agent class: Cytotoxic chemotherapy — antimetabolite / gemcitabine

RT site: (No RT site selected)

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *In the RT Interact app, select the agent shown but leave the RT site unselected. Assess how the app handles this incomplete input.*

Scenario S034

Agent: Nivolumab

Agent class: Immune checkpoint inhibitor — anti-PD-1

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: (Not selected)

Task: *In the RT Interact app, select the agent, RT site, and timing shown but leave the fractionation field unselected. Assess whether the app correctly prompts for the missing field.*

Multi-agent scenarios (S035–S036)

Scenario S035

Agent: Ipilimumab + Nivolumab

Agent class: Immune checkpoint inhibitor — dual ICI (anti-CTLA-4 + anti-PD-1)

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Select all agents listed and enter these inputs into RT Interact. Assess the output for clinical appropriateness, noting that combination-specific effects between agents may not be fully modelled.*

Scenario S036

Agent: Carboplatin + Paclitaxel

Agent class: Cytotoxic chemotherapy — platinum + taxane doublet

RT site: Lung — Central

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Select all agents listed and enter these inputs into RT Interact. Assess the output for clinical appropriateness, noting that combination-specific effects between agents may not be fully modelled.*

Timing-sensitive scenarios (S037–S039)

Scenario S037

Agent: Bevacizumab

Agent class: Anti-VEGF monoclonal antibody

RT site: Upper abdomen — Liver

Timing: Planned after RT

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact, paying particular attention to whether the timing and interval are handled appropriately and whether residual risk from recent therapy is reflected.*

Scenario S038

Agent: Capecitabine

Agent class: Cytotoxic chemotherapy — fluoropyrimidine

RT site: Pelvis — Rectal

Timing: Recent before RT

Timing interval: 14 days

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact, paying particular attention to whether the timing and interval are handled appropriately and whether residual risk from recent therapy is reflected.*

Scenario S039

Agent: Doxorubicin

Agent class: Cytotoxic chemotherapy — anthracycline (radiation recall agent)

RT site: Breast — Chest wall

Timing: Recent after RT

Timing interval: 21 days

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact, paying particular attention to whether the timing and interval are handled appropriately and whether residual risk from recent therapy is reflected.*

Fractionation-sensitive scenarios (S040–S041)

Scenario S040

Agent: Nivolumab

Agent class: Immune checkpoint inhibitor — anti-PD-1

RT site: Lung — Peripheral

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the fractionation category is appropriately reflected in the alert level and toxicity domains.*

Scenario S041

Agent: Palbociclib

Agent class: CDK4/6 inhibitor

RT site: Pelvis — Rectal

Timing: Concurrent

Fractionation: SABR/SBRT

Task: *Enter these inputs into RT Interact and assess whether the fractionation category is appropriately reflected in the alert level and toxicity domains.*

Site-specific interaction (S042)

Scenario S042

Agent: Trastuzumab

Agent class: HER2-directed monoclonal antibody (non-ADC)

RT site: Breast — Breast/chest wall + regional nodes

Timing: Concurrent

Fractionation: Conventional fractionation

Task: *Enter these inputs into RT Interact and assess whether the app output, including agent classification and toxicity domains, is clinically appropriate for this agent-site combination.*

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7. Known Limitations

The following limitations apply to the current prototype and are relevant to your content assessment.

Regulatory and Approval Status

- Not approved or cleared for clinical reliance in any jurisdiction.
- Not a medical device as currently constituted; no TGA, FDA, or CE process has been initiated.
- Intended solely for expert content review and quality improvement research.

Content Verification Incomplete

- ARTG/PBS verification incomplete: agent names, classifications, and availability have not been fully verified.
- eviQ verification incomplete: rules have not been cross-checked against current eviQ cancer treatment protocols.
- Expert consensus validation incomplete: alert levels, toxicity domains, and rationale text have not undergone prior formal consensus validation.

Clinical Scope Limitations

- No patient-specific factors: renal/hepatic function, performance status, comorbidities, and prior toxicity are not modelled.
- No prior RT or re-irradiation assessment.
- No exact dose or dose-constraint logic: alert levels are not tied to specific prescription doses or OAR constraints.
- No management recommendations (dose modifications, monitoring protocols, supportive care).
- Multi-agent combination effects not individually modelled in Phase 1.
- No disease-specific context: rules are not stratified by histology, stage, or intent.

Technical Limitations

- Reporting backend is mock/local only; no data is transmitted or persisted.
- Agent list is not exhaustive; many agents are not yet included.
- Rule engine uses categorical logic only; continuous dose variables are not modelled.
- Evidence links may become outdated; external references were current at time of authorship.

Data Privacy

- No patient information is collected, stored, or transmitted by the application.
- All queries and reports remain in the user's browser only.

8. Contact and Feedback

If you have questions about a scenario, the review process, or the prototype, please contact:

[PROJECT LEAD NAME] · [EMAIL ADDRESS] · [PHONE / PREFERRED CONTACT METHOD]

Please complete all assigned scenarios before the review deadline communicated in your invitation. If you are unable to complete all scenarios, please notify the project lead as soon as possible.